

Subject: Optimizing Web Scraping Strategy for Google News

Current Approach and Challenges: Our current web scraping approach encounters difficulties due to Google's security measures, leading to IP blocks and data corruption caused by CAPTCHA restrictions. The key risk factors include:

1. **Scraping News:** Parsing static HTML is straightforward, but excessive requests may still raise detection risks.
2. **Fetching Redirected URLs:** Multiple requests to Google News increase the likelihood of detection and blocking.
3. **Scraping Article Content:** Extracting content from redirected URLs can trigger CAPTCHA challenges, as noted in past project reviews.

Proposed Solutions to Mitigate Risks:

1. **Assign a Single Category per Person:**
 - Each team member scrapes only one category per day. However, this may not be feasible as the project requires multiple categories to be scraped.
 - Instead, data from multiple team members working on different categories can be compiled into a single shared document and reviewed collectively.
2. **Introduce Randomized Scraping Times:**
 - Avoid scraping at fixed times daily to prevent detection by Google's security systems.
 - Implement time variations to reduce the likelihood of triggering security measures.

API-Based Alternative for a Reliable Approach:

Instead of web scraping, using APIs can provide direct access to news data while reducing risks. However, API limitations must be considered:

1. **Use Third-Party APIs to Prevent Blocking:**
 - Implement third-party APIs like **SerpAPI** (<https://serpapi.com/google-news-api>) to fetch Google News data efficiently.
 - SerpAPI allows free **100 queries per month**, which can be optimized by assigning three categories per person daily while staying within the limit.

Action Plan:

1. Approve the structured approach of **one category per person per day** for sustainable scraping.
(New Word Document Not Possibel)
(or)
2. Evaluate **SerpAPI or similar services** for API-based data retrieval to ensure smooth and secure data collection.

Please review and approve the suggested strategy so we can proceed efficiently.